

Course Code: DSA201

Course Name: Data Structures and Algorithms

Semester: Fall 2026

Course Objectives:

Learn fundamental data structures and algorithms for problem solving.

UNIT 1: Complexity Analysis

- Time Complexity
- Space Complexity
- Big O Notation

UNIT 2: Linear Data Structures

- Arrays
- Linked Lists
- Stacks
- Queues

UNIT 3: Trees and Graphs

- Binary Trees
- BST
- Graph Traversal
- BFS and DFS

UNIT 4: Searching and Sorting

- Binary Search
- Merge Sort
- Quick Sort
- Heap Sort

UNIT 5: Dynamic Programming

- Memoization
- Knapsack
- LCS
- DP Problems

Suggested Books:

- Cormen – Introduction to Algorithms
- Narasimha Karumanchi – Data Structures